Curriculum Vitae
Mahdee Jodayree, Ph.D.

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Personal Website: Website Link   Google Scholar: Google Scholar

PROFESSIONAL SUMMARY

Ph.D. graduate in Computer Science from McMaster University, Canada, holding a Master's degree in Information Technology (computer networks) and double Bachelor's degrees in Mathematics and Computer Science, all earned in Canada. Research-focused scholar specializing in secure and distributed machine learning, verification-driven AI systems, and privacy-aware federated learning architectures for cloud–edge and IoT environments.

Proven independent research productivity with 13 peer-reviewed publications as first author, including six Q1 journal articles, two scholarly books, and five peer-reviewed conference papers, contributing to verification-driven learning systems, robust model generalization, and privacy-aware distributed training. Demonstrated expertise in designing cryptographic integrity mechanisms, adversarial resilient model pipelines, and reproducible large-scale experimentation frameworks.

Well positioned to contribute to high-impact, interdisciplinary research initiatives aligned with advanced artificial intelligence, secure distributed systems, and next-generation trustworthy machine learning infrastructure. **Canadian citizen, fully eligible to work in Canada.**

PUBLICATIONS

Q1 Journal Publications (First Author) — Published, Under Review

1. Jodayree, M., "Explainable Zero-Day Attack Detection in IoMT Using Transformer-based Time-Series Modeling", Scientific Reports — Q1
2. Jodayree, M., "A Systematic Literature Review of Artificial Intelligence Methods Applied to the Human Epidemic (COVID-19)", Applied Network Science — Q1
3. Jodayree, M., He, W., and Janicki, R., "Proof-Before-Train Federated Learning: ECC-Backed Zero-Leak Commitments and Trust-Aware Aggregation", Scientific Reports — Q1
4. Jodayree, M., "Uncertainty-Aware Hybrid Autoscaling with Bias-Corrected Deep Forecasting for Cost-Efficient SLA Preservation Across Cloud Workloads", Scientific Reports — Q1
5. Jodayree, M., "Federated Learning Across Schools to Predict Student Outcomes: Secured With Proof-Before-Train and Differential Privacy", Scientific Reports — Q1

6. Jodayree, M., Atashafrouz, M., Raeispour Rajabali, A., Jalalkamali, M., et al., “The Role of Supply Chain Finance (SCF) Platforms in Mitigating Financial Risk and Improving Resilience”, American Institute of Mathematical Sciences (AIMS) — Q2, DOI: 10.3934/jdg.2026015

Books

7. Jodayree, M., “Agritech — Focusing on Precision Agriculture, AI and ML Applications, Blockchain, Mobile Applications, and Digital Logistics”, Book Chapter

8. Jodayree, M., “Prevention of Data Poisoning Attacks in Federated Machine Learning — Persian Edition”

Conference Papers

9. Jodayree, M., He, W., and Janicki, R. An Encrypted Forensic Method for Preventing Data Poisoning Attacks in Federated Learning. FSDM 2023, Matsue, Japan.

10. Jodayree, M., He, W., and Janicki, R. Preventing Text Data Poisoning Attacks in Federated Machine Learning by an Encrypted Verification Key. IJCRS 2023, Kraków, Poland.

11. Jodayree, M., He, W., and Janicki, R. Preventing Image Data Poisoning Attacks in Federated Machine Learning by an Encrypted Verification Key. KES 2023, Athens, Greece.

12. Jodayree, M., and Janicki, R. A Predictive Resources Management for Clouds. FLINS 2020, Cologne, Germany.

13. Jodayree, M., Abaza, M., and Tan, Q. A Predictive Workload Balancing Algorithm in Cloud Services. Procedia Computer Science, 159, 902–912, 2019, Budapest, Hungary.

EDUCATION AND RESEARCH EXPERIENCE

McMaster University 2018 – 2024

Degree: Ph.D. **Hamilton, Ontario**

Area of Study: Computer Science

Dissertation: Distributed Machine Learning and Secure Federated Learning for IoT and Healthcare Systems

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Athabasca University 2016 – 2018

Degree: Master of Science (M.Sc.) **Athabasca, Alberta**

Area of Study: Information Systems

Thesis: Machine-learning–driven predictive load balancing for cloud services

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Carleton University 2005 – 2013

Degree: Bachelor of Science (B.Sc.) **Ottawa, Ontario**

Area of Study: Computer Science and Mathematics (Dual Degree)

Ontario Secondary School Diploma (OSSD)

J.S. Woodsworth Secondary School, Ontario, Canada

TEACHING AND ACADEMIC EXPERIENCE

Instructor & Researcher & PhD Student **Jan 2021 – May 2025**

McMaster University **Hamilton, ON**

Teaching Experience – Graduate and Undergraduate Courses

- Taught undergraduate Computer Science courses in programming (Python, Java, C, C++, C#, MATLAB, R, TensorFlow, PyTorch, Scikit-learn, Keras, JAX), data structures and algorithms, discrete mathematics, machine learning, data analytics, cybersecurity, computer networks, cloud computing, computer architecture, SDLC, and capstone projects.
- Designed and delivered hands-on, experiential labs integrating real-world computing environments and applied problem-solving.
- Delivered in-person, online, and hybrid instruction using Canvas, Moodle, institutional learning management systems (LMS), Microsoft Teams, and Zoom, ensuring regular and substantive student engagement.
- Guided undergraduate and graduate student teams, startup founders, and research groups in preparing successful Mitacs funding applications, supporting technical project design, cloud architecture proposals, and innovation objectives.
- Led workshops and training sessions on DevOps, source control, cloud-native development, and infrastructure-as-code (Terraform), using active-learning approaches to develop practical, job-ready skills.

CERTIFICATIONS

Teaching and Learning Certificate from MacPherson Institute, McMaster University

Courses Completed:

- Microsoft Certified: Azure Fundamentals AZ-900
- Microsoft Certified: Azure Data Fundamentals DP-900
- CompTIA A+ Certification: Remote Support Technician

TECHNICAL SKILLS

Artificial intelligence & machine learning

Federated learning, Distributed machine learning, Privacy-preserving learning, Verification-driven learning systems, Trustworthy AI, Robust model generalization, Adversarial machine learning, Differential privacy, Transformer architectures, Large language models (LLMs), Explainable AI (XAI), Time-series modeling, Representation learning, Model evaluation and validation frameworks, AI-driven cybersecurity, Intelligent threat detection

AI security & cybersecurity

Secure distributed systems, Cyber-physical systems security, Critical infrastructure protection, Industrial control systems (ICS) security, Operational technology (OT) security, SCADA security, Adversarial threat modeling, Intrusion detection systems, Security analytics, Network anomaly detection, Data poisoning detection and mitigation, Cryptographic integrity mechanisms, Secure aggregation protocols, Privacy-aware distributed training, Semantic consistency verification, Trust-aware learning architectures, Resilient model pipeline design

Machine learning frameworks & research tools

PyTorch, TensorFlow, Hugging Face Transformers, Scikit-learn, NumPy, Pandas, JAX, Reproducible experimentation frameworks, Large-scale model training pipelines, Experimental design and benchmarking, Security data analytics pipelines

Distributed systems & cloud-integrated research infrastructure

Cloud computing architectures, Cloud–edge computing, Edge AI systems, Scalable computing environments, Containerized ML pipelines, Virtualized distributed systems, Performance modeling, Resource management in large-scale systems, Distributed AI infrastructure

DevOps & experimentation infrastructure

Docker, Kubernetes, CI/CD pipelines, Infrastructure as Code (Terraform), Automated experimentation pipelines, Version-controlled research workflows, Reproducible ML experimentation environments

Programming languages

Python, C, C++, Java, C#, MATLAB, R, Bash

Data & systems

SQL, Database systems, Data engineering for ML pipelines, System-level performance analysis, Computational modeling for distributed environments, Data processing for security analytics